5 Electric Energy: Consumption and Conservation

You have noticed a situation where electricity is wasted in homes, haven't you?

Do you waste electricity?

Which are the appliances in houses that operate using electricity?

- Tube light
- Fan
- Electric kettle
- Electric iron
-

Fig. 5.1

Are all electrical appliances used for the same purpose?

These appliances convert electric energy into different forms of energy.

What forms of energy are produced when a mixie operates? Write them down.

- Mechanical energy
- Sound energy
-

Isn't the electric energy we supplied that is transformed into various forms when the mixie is operated?

However, the useful form of energy into which an appliance mainly converts electric energy is considered the effect of electric current in that appliance.

Observe the appliances in figure 5.1 and the appliances already listed. Based on their function, identify the effect of electric current utilised in each and complete table 5.1.

Name of the appliance	Function	Energy conversion	Effect of electric current
Electric kettle	Heats water	Electric energy → Heat energy	Heating effect
Mixie			
Electric iron			

Table 5.1

From the table, you may have understood that electric current can produce various effects. Which of these appliances produce heat?

- Electric iron
-

Heating Effect of Electric Current

The conversion of electric energy mainly into heat energy is the heating effect of electric current. Appliances that utilise this effect are electric heating appliances.

Let's do an experiment.

Connect a 5 cm long nichrome wire to a 9 V battery as shown in figure 5.2.

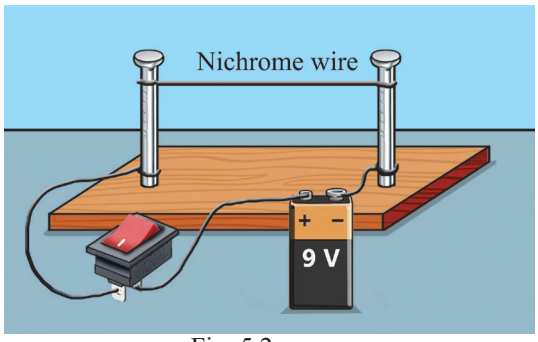


Fig. 5.2

- Switch on and observe the nichrome wire. What change do you notice?

When current passes through the nichrome wire, heat energy is produced.

The process of production of heat, when electricity flows through a conductor is Joule heating or Ohmic heating. The part used to produce heat energy in heating appliances is the heating element.

Observe the pictures of the given electric heating appliances.

Electric heater

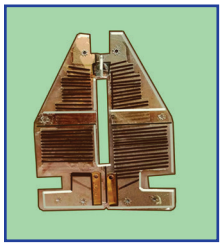

Heating coil in an iron box

Fig. 5.3

You may have noticed that the part common to all these appliances is the heating coil (heating element).

The heating coil is made of an alloy, nichrome.

What are the characteristics of nichrome that make it suitable for a heating element?

- High oxidation resistance (oxidation resistance is the ability of a material to resist corrosion due to contact with oxygen or other oxidisers at high temperature).
- Ability to provide heat energy for a long time in a red hot state.
- High resistivity (due to this property even a nichrome wire of short length can provide sufficient resistance)
-

Resistivity of certain materials

Material	Resistivity (Ωm)
Silver	1.59×10^{-8}
Copper	1.68×10^{-8}
Aluminium	2.65×10^{-8}
Tungsten	5.6×10^{-8}
Iron	9.71×10^{-8}
Lead	22×10^{-8}
Nichrome (Ni, Cr, Fe)	100×10^{-8}

We know that all materials do not conduct electricity in the same way.

Resistivity is one of the intrinsic properties of a material that mainly determines the resistance which is the characteristic of a conductor. Resistivity is different for each material.

The resistance of a conductor with an area of cross section of one square metre and a length of one metre is the resistivity of the material of the conductor. It is denoted by the Greek letter ρ (rho). Its unit is ohmmetre (Ωm). You have learnt in standard IX that the factors which influence the resistance of a conductor are length of the conductor (ℓ), area of cross section (A) and the nature of the material (ρ). The relation of resistance (R) with these factors is given by the equation $R = \frac{\rho \ell}{A}$.

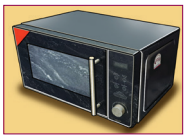
Conductivity is the ability of a material to conduct electricity. This is the reciprocal of resistivity.

Heating appliances that operate without heating coils are also available. Examples are induction cooker, microwave oven, etc.

Induction Cooker

The induction cooker works on the principle that a continuously varying magnetic field produces heat in a magnetic substance.

Microwave Oven



Only foods with water content can be cooked using a microwave oven. Foods are cooked by converting microwave into heat energy.

What are the factors that influence the heat produced in a conductor carrying current?

Let's do an activity.

Construct a circuit which consists of a 5 cm long nichrome wire, 6 V battery eliminator, etc., as shown in figure 5.4. Take 20 mL of water in a boiling tube. Immerse the nichrome wire and a thermometer into it. Adjust the rheostat to vary the current. Record the temperature in table 5.2 every two minutes. Repeat the experiment by replacing the water in the boiling tube each time.

Temperature of water before starting the experiment = ...

Current (A)	Temperature (°C)
0.5	
1	
1.5	

Table 5.2

PhET→ Resistance in a wire.

On analysing table 5.2, didn't you understand that the temperature of water increases with increase in current? In this experiment a fixed quantity of water (20 mL) is heated by passing current. To increase the temperature of a fixed quantity of water more heat is required. Hence as the quantity of heat received by the water increases, the temperature also increases. Therefore, it is understood that when the current increases heat produced also increases.

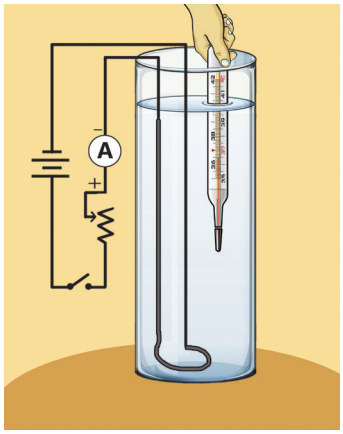


Fig. 5.4

Adjust the rheostat to make the current 1 A. In table 5.3 record the temperature of water, every 2 minutes.

Time (in minutes)	Temperature (°C)
0	
2	
4	
6	

Table 5.3

Didn't the temperature increase because the current flowed for a longer time? From the experiment, you may have understood the relation between the heat produced and the time of flow of current.

- Are there other factors that influence the heat produced in a conductor carrying current? Let's examine.

Take nichrome wire and aluminium wire of equal length (5 cm) and equal area of cross section. Using them make a circuit as shown in the figure. Pass current through the circuit for about 2 minutes using a 9 V, 50 W eliminator. Place a thin paper on both aluminium and nichrome wire. What do you observe?

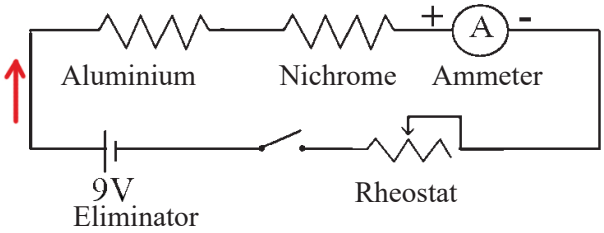


Fig. 5.5

- Which conductor becomes hotter?
- The magnitude of current in the nichrome wire and aluminium wire is

  (the same / different).
- The resistance of nichrome wire and aluminium wire is

  (the same / different).

Didn't the nichrome wire become hotter because its resistance is greater than that of the aluminium wire?

Resistance is another factor that influences the quantity of heat produced in a conductor carrying current.

From the experiments conducted, note down the three factors that influence the quantity of heat produced in a current carrying conductor :

- Current (I)
-
-

The scientist, James Prescott Joule discovered the relation between the heat produced in an electric conductor and the factors influencing the quantity of heat.

Joule's law

The quantity of heat produced in a current carrying conductor is directly proportional to the square of the current (I^2) the resistance of the conductor (R) and the time (t) for which the current flows. This is Joule's law.

If a current I flows through a conductor having resistance R for a time t, according to Joule's law, the quantity of heat produced will be $H = I^2Rt$. Its SI unit is joule (J).

- A current of 0.5 A flows through a conductor with 100 Ω resistance for 5 minutes.

(a) What will be the quantity of heat produced?

(b) If a current of 1 A flows through this conductor for 5 minutes, what will be the quantity of heat produced?

a) $R = 100\ \Omega$

$I = 0.5\ A$

$t = 5 \times 60\ s$

$H = ?$

$H = I^2Rt$

$= 0.5 \times 0.5 \times 100 \times 5 \times 60\ J$

$= 7500\ J$

b) $I = 1\ A,\ H = I^2Rt$

$H = 1 \times 1 \times 100 \times 5 \times 60\ J$

$= 30000\ J$

When the current was doubled, the heat produced in the conductor became 4 times. If so, what happens to the heat produced if the current is halved?

(halved / quartered)

Aren't electrical appliances commonly used in houses operating at 230 V?

- If the resistance of a heating appliance operating at 230 V is 920 Ω, what is the quantity of heat produced in 10 minutes?

The value of current is not given in this numerical problem. In this situation, can we formulate other equations to calculate the quantity of heat?

According to Ohm's law,

$$V = I \times R$$

$$\text{Hence } I = \frac{V}{R}$$

On substituting the value of I in the equation $H = I^2 Rt$

$$H = \left(\frac{V}{R}\right)^2 Rt$$

$$\text{So, we get } H = \frac{V^2 t}{R}$$

Similarly, if we substitute $R = \frac{V}{I}$ in the equation $H = I^2 Rt$

$$\text{we get } H = VIt$$

$$H = I^2 Rt = \frac{V^2 t}{R} = VIt$$

Here

$$V = 230 \text{ V}, \; R = 920\ \Omega, \; t = 10 \times 60 \text{ s},$$

$$H = \frac{V^2 t}{R}$$

$$= \frac{230 \times 230}{920} \times 10 \times 60 = 34500 \text{ J}$$

If 7500 J of heat is produced in 30 s in a conductor having 250 Ω resistance, what is the current through the conductor?

Calculate the quantity of heat produced if 2 A current flows for 10 minutes through a heating coil of an electric kettle having 100 Ω resistance?

A potential difference of 230 V is applied across a circuit for 5 minutes. If the resistance in the circuit is as given below, calculate the current and heat in each case.

(a) 115 Ω (b) 230 Ω

Observe the circuits given below. Calculate the quantity of heat produced in each, in 5 minutes.

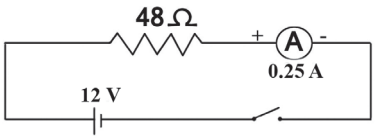


Fig. 5.6 (a)

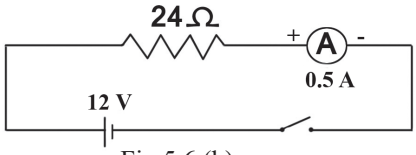


Fig 5.6 (b)

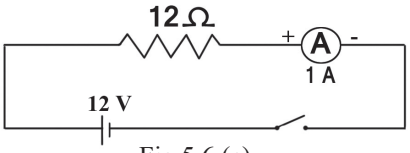


Fig 5.6 (c)

Heat produced in circuit 5.6 (a)	Heat produced in circuit 5.6 (b)	Heat produced in circuit 5.6 (c)
$H = I^2Rt$ $= 0.25 \times 0.25 \times 48 \times 300$ $= 900$ J	$H = I^2Rt$	$H = I^2Rt$

Table 5.4

Analysing the completed table, answer the following questions.

- What is the difference in the quantity of heat in circuits 5.6 (a) and 5.6 (b)?
- Why is there a difference in the quantity of heat produced in these circuits?
- What is the resistance of the circuit in figure 5.6 (c)?
- What is the total heat produced in the circuit in figure 5.6 (c)?
- Less heat is produced in the resistor of higher resistance in the circuit in figure 5.6 (a). Why?
- Among these circuits, in which resistor is the maximum quantity of heat produced?

What are your inferences? Record them in the science diary.

If voltage is constant, when the resistance in the circuit is decreased, the current increases. Hence the quantity of heat produced increases.

If the voltage remains constant, what change will occur in the quantity of heat produced when the resistance varies? Explain based on Joule's law.

Calculate the quantity of heat produced if 2 A current flows through an electric heater operating at 230 V for 15 minutes. What is the resistance of the heater?

An electric iron operating at a potential difference of 230 V has a heating coil of resistance 100 Ω. If this electric iron operates for half an hour, how much electric energy will be converted to heat energy? What is the current through the electric iron?

A heating appliance having a resistance of 92 Ω operates at 230 V. Using various equations, calculate the heat produced by the appliance in 14 minutes and write them down in the table below. What resistance is needed to double the heat energy?

$H = I^2Rt$	$H = \frac{V^2 t}{R}$	$H = VIt$
$V = 230$ V $R = 92\ \Omega$ $t = 14$ minute $= 14 \times 60$ s $= 840$ s $I = \frac{V}{R} = \frac{230\ V}{92\ \Omega} = 2.5$ A $H = I^2 Rt$ $= (2.5)^2 \times 92 \times 840$ J $= 483000$ J		

Table 5.5

Is there any other method to find the quantity of heat produced by heating appliances?

Observe the values given on the label of the appliance in figure 5.7. What does the marking 750 W indicate?

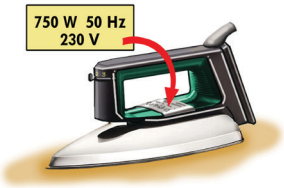


Fig 5.7

Energy conversion occurs when this appliance operates. That is, work is done. The quantity of work done per unit time is power. We have already learned in standard IX that power is the rate of work based on time.

- How can we find the rate of work of an electrical appliance based on time?

Electric Power

Power is the quantity of work done by an electrical appliance per unit time.

The work done by an electrical appliance is the conversion of electric energy into another form of energy.

The function of electric heating appliances is to produce heat. You have learnt that the quantity of heat produced $H = I^2 Rt$.

$$\text{Work} = \text{Quantity of heat produced}$$

$$W = H = I^2 Rt$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}}$$

$$P = \frac{W}{t} = \frac{H}{t}$$

$$P = \frac{I^2 Rt}{t}$$

$$P = I^2 R$$

According to Ohm's law $I = \frac{V}{R}$

Then, $P = \ldots\ldots\ldots\ldots\ldots\ldots$

We know that $R = \frac{V}{I}$

Hence $P = \ldots\ldots\ldots\ldots\ldots\ldots$

We will get the equations $P = \frac{V^2}{R}$, $P = VI$ as well

$$P = I^2 R = \frac{V^2}{R} = VI$$

If we substitute the value of P in the equations used to calculate heat, $H = I^2Rt$, what will be the value of H?

We get $H = Pt$

It is understood that the quantity of heat produced in a given time can be found if the power of an appliance is known.

The power of each electric appliance is marked on it. Complete the table given below.

Device	Power (P) W	Energy consumed in 1 second ($Pt = P \times 1$)
LED Bulb	9	$= 9 \times 1 = 9$ J
Projector	100	
Fan	60	
Laptop	40	

Table 5.6

While labelling the power of an appliance, the voltage at which it is to be operated is also indicated. In our country, the power obtained while operating at 230 V is labelled on electrical appliances. If the appliance operates above or below 230 V, the power will vary accordingly. That is, if the voltage increases, the power increases and if the voltage decreases, the power also decreases. This may lead to the damage of appliances. (There are also some appliances that operate at 400 V).

Countries design electrical appliances and power according to their power supply voltage. In all electrical appliances, the voltage required for it and its power will be marked on it. The indicated power is available from an appliance only when it operates at the specified voltage.

Power : in watt (W) and volt ampere (VA)

Both are used to indicate power, but there is a small difference.

The actual power consumed by electrical appliances is recorded in watt (W). For example, bulb, electric iron, etc.

But the power that can be supplied by electric power generating devices is given in volt ampere (VA). For example, UPS, solar panel, etc.

1 kW = 1000 W

1 kVA = 1000 VA

If the voltage is constant, according to $P = \frac{V^2}{R}$, appliances with higher resistance will have lower power, and those with lower resistance will have higher power. While solving numerical problems related to household appliances, it is more desirable to use the formula $P = \frac{V^2}{R}$.

In some foreign countries, the supply voltage is 110 V.

At what applied voltage will an electric heating appliance marked 800 W, 240 V produce a power of 200 W?

Calculate the quantity of heat produced when a heating appliance of power 500 W operates for five minutes.

$P = 500$ W

$t = 5$ min

$= 5 \times 60 = 300$ s

$H = Pt = 500 \times 300$ J

$= 150000$ J

How much heat energy is produced when an electric heater with 600 W power operates for 20 minutes?

How much heat energy will be produced if a heating appliance with 250 W power operates for 10 minutes?

An electric heater produced 450000 J of heat energy when operated for 5 minutes. What would be the power of this heater?

On a heating appliance it is marked 529 W, 230 V.

a) What do each of these mean?

b) What would be the power of this appliance if 100 V is supplied?

Don't we use many types of electrical appliances in our daily lives? If so, let's see how the quantity of electric energy that we use, is calculated.

Watt - Hour Meter

Fig. 5.8

The quantity of electric energy consumed in houses can be measured directly by a watt hour meter connected to our household electric circuit. In this, electric energy is measured in kilowatt hour (kWh) units. This is the commercial unit of electric energy.

How much electric energy is one kilowatt hour or one unit?

Power is expressed in watt, time in second, and energy in joule. The energy consumed if a 9 W bulb operates for one hour daily for 30 days is $E = Pt = 9 \times 30 \times 3600$ J = 972000 J (nine lakh seventy two thousand joules). So, if we consider the monthly energy consumption of all the appliances in a house, the total would be very large, making it inconvenient to record. Therefore, energy is calculated in kilowatt hour by measuring power in kilowatt and time in hour.

- Calculate the energy consumed by a 1 kW power appliance in one hour.

$$E = Pt = 1\ kW \times 1\ h$$
$$= 1\ kWh$$

One kilowatt hour or one unit of electric energy is the energy consumed by an appliance with 1000 W (1 kW) power in one hour.

Let's see how many joules of energy is one unit of electric energy.

$$1000\ W = 1\ kW$$

Energy consumed, $E = H = Pt$

$P = 1000\ W,\ t = 1\ h = 60 \times 60\ s$

$H = 1000 \times 60 \times 60\ J$

$= 3600000\ J = 3.6 \times 10^6\ J$

- How much heat energy will be produced when a heating appliance of 60 W operates for one hour?

$H = Pt$

$= 60 \times 1 \times 60 \times 60\ J$

$= 216000\ J$

The power and operating time of some appliances are given in the table. Complete the table.

Power of device (P) W	**Time (t) h**	**Electric energy spent (Pt) J**	**Energy utilised in unit (kWh) $= \frac{\text{Power in watt}}{1000} \times \text{hour}$**
1000	1 hour	1000 × 60 × 60 = 3600000 J	1 kWh = 1 unit
2000	1 hour		
500	1 hour		
500	2 hour		

Table 5.7

Calculate the electric energy consumed if a 500 W grinder and a 600 W electric iron each operate for 2 hours.

TOD billing [Time Of the Day billing]

TOD billing is an innovative billing system introduced by KSEB to control electricity consumption during peak hours (6 pm to 10 pm). According to this, a day is divided into three time zones for the calculation of energy consumption. From 6 am to 6 pm, the billing rate is minimum. The time zone from 6 pm to 10 pm is considered the peak time with the highest charge. The time zone from 10 pm to 6 am is considered for normal charge. Therefore, please ensure that high power appliances like electric iron, motor pump, refrigerator, grinder, AC, etc., are operated during the minimum charge time zone.

Energy Star Rating

Normally ratings range from 1 to 5 stars. 5 stars represent the highest energy efficiency.

The energy consumption of higher star rated appliances is low when compared to similar low star rated appliances. Bureau of Energy Efficiency (BEE), an agency under Government of India assigns the star values.

The advantages of choosing higher star rated appliances :

- Lower electricity bills : By consuming less energy, you can reduce your monthly electricity costs.
- Environmental impact : Energy efficient appliances help to reduce greenhouse gas emissions.
- Government benefits : Many governments offer rebates or incentives to purchase higher star rated products.

$$\text{Total power in watt} = 500\ \text{W} + 600\ \text{W}$$
$$= 1100\ \text{W}$$
$$\text{Time} = 2\ \text{h}$$
$$\text{Electric energy (in kilowatt hour)} = \frac{\text{Power in watt} \times \text{time in hour}}{1000}$$
$$= \frac{1100\ \text{W} \times 2\ \text{h}}{1000}$$
$$= 2.2\ \text{kWh}$$
$$= 2.2\ \text{unit}$$

Using this formula, we can calculate the monthly consumption of electricity.

Considering the total power of the appliances used and their operating time, calculate the daily electricity consumption in your house. Similarly, calculate the monthly consumption of electricity and present in class the main findings and suggestions based on the project report.

Let's see if all houses have the same electricity tariff.

Electricity charges are levied at a higher rate from consumers who use more electricity and at a lower rate from consumers who use less electricity. (Table 5.8, page 112)

- Now you have understood one of the reasons for the increased consumption of electricity in the situation depicted at the beginning of the lesson.

Have you noticed the energy consumption recorded in the electricity bill of your house? The amount in the electricity bill can be reduced by lowering electricity consumption.

- What can we do to reduce electricity consumption at home?
  - Use energy efficient electrical appliances.
  - Turn off switches immediately after use.
  - Use LED tubes and bulbs.
  - Choose the size of the fan according to the size of the room.
  - Use BLDC fans. [BLDC – Brush Less Direct Current]
  -

Further activities to reduce energy consumption can be implemented under the leadership of the School Energy Club in collaboration with the Energy Management Centre (EMC).

Energy Crisis

Despite having many small scale electricity projects in addition to large power stations, the electricity they produce is not sufficient for our needs. Though the demand for energy has increased many folds, production has not sufficiently increased. This situation is energy crisis. This is why sometimes power cuts and load shedding are to be implemented.

'Energy crisis is the increase in demand for energy and the decrease in availability.'

There are hydroelectric power plants, thermal power plants, nuclear power plants, etc., to produce electricity on a large scale. Yet, it is not easy to increase the electricity production. Do large scale electricity generating power plants cause environmental impacts? Discuss

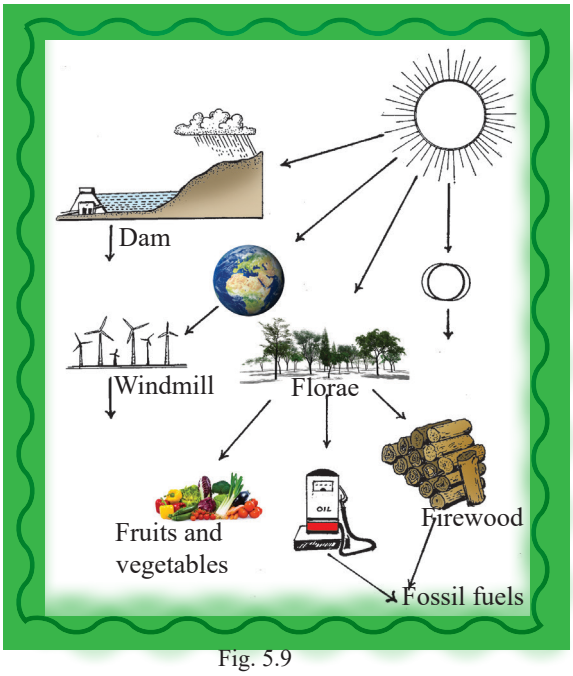


Fig. 5.9

Is there any way to produce electricity without causing pollution?

Observe figure 5.9. You know that the sun is the basis of all energy on the earth.

List the devices that utilise solar energy.

- Solar water heater
- Solar cooker
-

The solar power plant at Cochin International Airport : This energy is used for all airport operations. The remaining is given to KSEB. This green energy initiative received the United Nations' highest environmental award in 2018.

Solar cells are devices that convert solar energy into electric energy. Only negligible voltage and current can be obtained from a single cell. Therefore, a solar panel is constructed by arranging many solar cells suitably to provide electricity as per the requirement. The electricity obtained from such panels can be stored in a storage battery and utilised when needed. Or it can be directly supplied to the electricity distribution agencies. Solar panels are used for energy requirements in artificial satellites. In addition to small scale requirements, large scale solar power plants are also in operation. Cochin International Airport Limited (CIAL) in Nedumbassery, Kerala, produces electricity required for its entire operations from such a solar power plant.

Cochin International Airport is the world's first airport to operate entirely on solar power.

Solar panels can be installed on rooftops of houses and other places receiving sunlight. The Rooftop Power Project is a scheme jointly initiated by the Central Government and KSEB for this purpose.

Rooftop Power Project

List the benefits of installing solar panels.

- Can produce electricity required for homes.
- Can reduce environmental pollution.

Fig. 5.10

- Transmission and distribution losses are minimised as electricity is generated at the place of consumption.

Let's work together for a better tomorrow by implementing such innovative activities and taking measures to reduce the energy crisis.

It is understood that atmospheric pollution is reduced when solar panels are used. We know that the quantity of greenhouse gases in the atmosphere is increasing daily. An increase in such gases cause global warming, resulting in climate change. Each of our interventions in this matter is very

important. Global warming can be reduced by reducing the production of major greenhouse gases like carbon dioxide, methane etc. We need activities that can reduce the quantity of carbon and carbon compounds emitted by individuals, organisations, and products.

Carbon footprint

Greenhouse gases are directly and indirectly emitted by individuals, families, organizations, events, services, products, etc. The quantity of such emitted greenhouse gases, converted and expressed as equivalent to the measure of carbon dioxide, is carbon footprint.

Fig. 5.11

Each individual should strive to reduce his/her personal carbon footprint. This can be achieved by paying attention to daily activities. How one travels, what food one eats, what clothes one uses, and how much waste is generated are all important.

What can be done to reduce carbon footprint?

- Reduce domestic energy consumption.
- Avoid wasting food.
- Use public transport.
- Reduce waste by utilising reusable products.
- Educate society about reducing carbon footprint.
- Perform energy consuming activities in an energy saving manner.

Prepare a seminar paper on how to reduce individual carbon footprint and present it in class.

'Reducing atmospheric pollution and conserving energy is everyone's duty. Energy conservation is equivalent to energy production.'

'Energy is precious, don't waste it !'

Let's Assess

1. An electric heater operating at 230 V draws 2 A current.

   a) What is the resistance of the heater?

   b) Calculate the heat produced when this heater operates for 10 minutes.

   c) What is the power of this appliance?

2. a) A heating appliance operating at 230 V supply consumes 2 A current. What is the quantity of heat produced in five minutes?

   b) What is the energy consumed by this appliance in five minutes while operating at 115 V?

3. In a house, a 500 W electric iron operates for one hour, two 40 W fans for 8 hours, and five 12 W LED bulbs for 10 hours. Calculate in unit the energy consumption per day in that house.

The tariff of electricity for domestic consumers in Kerala is given below (subject to change).

Monthly Consumption kWh	Fixed Charge Rs./Consumer /Month		Energy Charge per unit	Remarks
	Single Phase	Three Phase		
0 - 40	Nil		1.50	This Rate is Applicable only for BPL Category with Connected load of And below 1000 W
0 - 50	50	130	3.35	Telescopic
51 - 100	85	175	4.25	
101 - 150	105	205	5.35	
151 - 200	140	215	7.2	
201 - 250	160	235	8.5	
0 - 300	220	240	6.75	Non-Telescopic
0 - 350	240	250	7.6	
0 - 400	260	260	7.95	
0 - 500	286	285	8.25	
Above 500	310	310	9.2	
0 - 50	40	100	1.50	BPL family having Cancer Patients or Permanently Disabled person as family members due to polio or accidents with Connected load of And below 2000 W
51 - 100	65	140		

Source : Official website of KSEB

Table 5.8

4. In a house, a 600 W grinder operates for one hour, three 60 W fans for six hours, and ten 10 W LED bulbs for 10 hours daily.

   a) What will be the electric energy consumed in units per month?

   b) If the appliances operate in the same way, calculate the electricity bill for two months in that house. (Use the tariff given in table 5.8)

5. Two heaters A and B operate at 230 V. Heater A draws 2 A current and heater B draws 2.5 A current.

   a) Calculate the power of the heaters in figure 5.12 (a) and (b).

   b) Which heater has higher resistance?

   c) If both heaters operate for 5 minutes each, which heater will produce more heat?

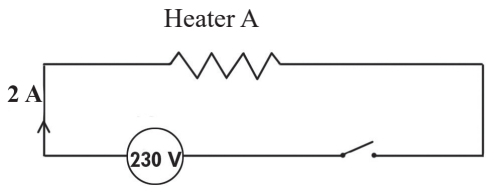


Fig. 5.12 (a)

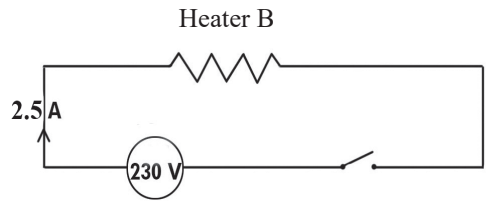


Fig. 5.12 (b)

6. Which of the given statements is correct with regard to the heating element?

   a) Low melting point

   b) High resistivity

   c) Ability to remain in red hot state

   d) Low oxidation resistance

   e) High melting point

   f) High oxidation resistance

   i. abfc ii. abfe iii. bdfe iv. bcef

7. Two bulbs, 230 V, 40 W and 230 V, 60 W are arranged as shown in the figures.

   If 230 V is applied, which bulb will be brighter in each circuit? Explain the reason.

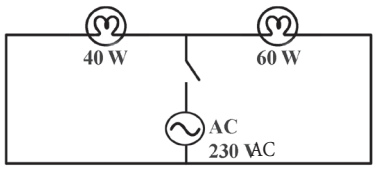


Fig. 5.13 (a)

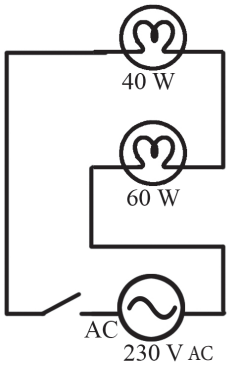


Fig. 5.13 (b)

8. To reduce electricity consumption in our houses, we can maximise the use of sunlight during the day. Do you agree with this statement? Explain how.

9. In the context of the energy crisis, write down any two suggestions that can be implemented to reduce energy consumption in newly constructed houses.

10. What is the necessity for reducing carbon footprint?

Extended activities

1. Observe the electricity bill of your house and find out the monthly electricity consumption. Present activities that can be implemented in your house to reduce electricity consumption in a seminar in the Energy Club. Also, prepare necessary posters for this.

2. List the activities that the School Science Club intends to undertake to reduce the school's carbon footprint.